

Causal inference

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Introduction

- We study methods for inference on *causal effects*.
- Starting point: *identification*
 - A set of assumptions linking the *estimand* to observable data
 - Allows expressing the causal quantity of interest as a function of the data distribution
- Key distinction:
 - Identification $\neq$ inference
- Once identification holds:
 - Inference can be *Frequentist* or *Bayesian*

Introduction

- **Identification:**
 - Can the causal effect be written as a function of the observable data distribution?
 - Requires a set of assumptions
- Once identified:
 - We can proceed with statistical inference
- Conclusion:
 - Identification is logically prior to the choice of inferential framework

Identification

- Goal:
 - Identify the *causal (treatment) effect* of a regressor on an outcome
- Examples:
 - Effect of a labor training program on earnings or unemployment
 - Price elasticity of demand
- Framework:
 - *Potential outcomes* framework (Rubin, 1974)
 - Defines *counterfactuals*: what would happen under alternative treatments
 - Also known as the *Rubin causal model* (Holland, 1986)

Identification

- Binary treatment:
 - $D_i = 1$ (treated), $D_i = 0$ (control)
- Potential outcomes:
 - $Y_i(1)$: outcome under treatment
 - $Y_i(0)$: outcome under control
- Individual treatment effect:

$$\tau_i = Y_i(1) - Y_i(0)$$

Identification

- Continuous treatment:
 - $Y_i(x_s)$: outcome if $X_s = x_s$ (Imbens, 2014)
- Discrete change in treatment:

$$\beta_{si} = Y_i(x'_s) - Y_i(x_s)$$

- Example: effect of a 10% increase in price on demand
- Infinitesimal change (marginal effect):

$$\beta_{si} = \frac{\partial Y_i(x_s)}{\partial x_s}$$

Identification

- **Fundamental problem of causal inference** (Holland, 1986):
 - For each unit i :
 - We observe either $Y_i(1)$ or $Y_i(0)$, but *never both*
- Missing data problem:
 - The counterfactual outcome is unobserved
- Implication:
 - Causal effects require comparisons across units

Randomized controlled trial (RCT)

Two of the most relevant estimands in the causal inference literature are the average treatment effect (ATE),

$$\text{ATE} = \mathbb{E}[Y_i(1) - Y_i(0)],$$

and the average treatment effect on the treated (ATT),

$$\text{ATT} = \mathbb{E}[Y_i(1) - Y_i(0) \mid D_i = 1].$$

Note that the observed mean difference between treated and untreated units can be decomposed as:

$$\begin{aligned} \mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0] &= \underbrace{\mathbb{E}[Y_i(1) - Y_i(0) \mid D_i = 1]}_{\text{ATT}} \\ &\quad + \underbrace{\left(\mathbb{E}[Y_i(0) \mid D_i = 1] - \mathbb{E}[Y_i(0) \mid D_i = 0] \right)}_{\text{Selection Bias}}. \end{aligned}$$

Selection Bias: Evidence from Observational Data

Health and demographic characteristics of insured and uninsured couples in the NHIS

	Husbands			Wives		
	Some HI (1)	No HI (2)	Difference (3)	Some HI (4)	No HI (5)	Difference (6)
A. Health						
Health index	4.01 [.93]	3.70 [1.01]	.31 (.03)	4.02 [.92]	3.62 [1.01]	.39 (.04)
B. Characteristics						
Nonwhite	.16	.17	-.01 (.01)	.15	.17	-.02 (.01)
Age	43.98	41.26	2.71 (.29)	42.24	39.62	2.62 (.30)
Education	14.31	11.56	2.74 (.10)	14.44	11.80	2.64 (.11)
Family size	3.50	3.98	-.47 (.05)	3.49	3.93	-.43 (.05)
Employed	.92	.85	.07 (.01)	.77	.56	.21 (.02)
Family income	106,467	45,656	60,810 (1,355)	106,212	46,385	59,828 (1,406)
Sample size	8,114	1,281		8,264	1,131	

Notes: This table reports average characteristics for insured and uninsured married couples in the 2009 National Health Interview Survey (NHIS). Columns (1), (2), (4), and (5) show average characteristics of the group of individuals specified by the column heading. Columns (3) and (6) report the difference between the average characteristic for individuals with and without health insurance (HI). Standard deviations are in brackets; standard errors are reported in parentheses.

Source: *Mastering 'Metrics: The Path from Cause to Effect*, Angrist and Pischke (2015), Table 1.1.

Randomized controlled trial (RCT)

Note that if we assume that *assignment to treatment is random*,

$$\{Y_i(1), Y_i(0)\} \perp D_i,$$

then the selection bias becomes zero,

$$\mathbb{E}[Y_i(0) \mid D_i = 1] - \mathbb{E}[Y_i(0) \mid D_i = 0] = 0,$$

which means there is no selection bias under random assignment.

Moreover, under random assignment, ATT equals ATE:

$$\tau = \underbrace{\mathbb{E}[Y_i(1) \mid D_i = 1] - \mathbb{E}[Y_i(0) \mid D_i = 1]}_{\text{ATT}} = \underbrace{\mathbb{E}[Y_i(1) - Y_i(0)]}_{\text{ATE}}.$$

Consequently,

$$\tau = \mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0].$$

Selection Bias

- **Selection bias:**

- Difference in untreated potential outcomes across groups

Scenario	$\mathbb{E}[Y(0) \mid D = 1]$	$\mathbb{E}[Y(0) \mid D = 0]$	Bias
Training Program	80	70	+10
Remedial Education	60	70	-10

- Positive bias: treated units would do better anyway
- Negative bias: treated units would do worse anyway

Randomized Controlled Trials (RCTs)

- Key feature:
 - *Random assignment* to treatment
- Identification requires:
- **1. Overlap**
 - $0 < P(D_i = 1) < 1$
 - Every unit can be treated or untreated
- **2. SUTVA (Stable Unit Treatment Value Assumption)**
 - *No interference*: outcomes do not depend on others' treatment
 - *Consistency*: observed outcome matches the realized treatment

Randomized controlled trial (RCT)

- The causal mechanism can be represented in an RCT using *causal diagrams* (Pearl, 1995, Pearl and Mackenzie, 2018).
- *Directed Acyclic Graph (DAG)* is a graphical representation of a set of variables and their assumed causal relationships.

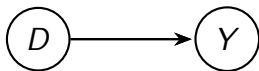


Figure 1: Directed Acyclic Graph (DAG) implied by a Randomized Controlled Trial (RCT).

Randomized controlled trial (RCT)

The counterfactual representation of the DAG, known as the *Single-World Intervention Graph* (SWIG)

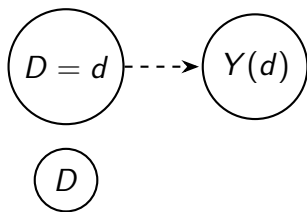


Figure 2: Single-World Intervention Graph (SWIG) for intervention $do(D = d)$: Y is replaced by the counterfactual $Y(d)$, and D is split into the fixed value $D = d$ and its natural value D . The dashed arrow indicates the fixed causal assignment.

Randomized Controlled Trials (RCTs)

- Observed outcome:

$$Y_i = [Y_i(1) - Y_i(0)]D_i + Y_i(0)$$

- Interpretation:
 - Equals $Y_i(0)$ if $D_i = 0$
 - Equals $Y_i(1)$ if $D_i = 1$
- Assume:
 - Constant treatment effect
 - Linear CEF: $\mathbb{E}[Y_i | D_i]$
- Linear representation (Angrist and Pischke, 2009):

$$Y_i = \underbrace{\beta_0}_{\mathbb{E}[Y_i(0)]} + \underbrace{\tau}_{\text{treatment effect}} D_i + \underbrace{\mu_i}_{Y_i(0) - \mathbb{E}[Y_i(0)]}$$

- τ : common treatment effect

Randomized Controlled Trials (RCTs)

- Under random assignment:

$$\mathbb{E}[\mu_i \mid D_i] = \mathbb{E}[Y_i(0) \mid D_i] - \mathbb{E}[Y_i(0)] = 0$$

- Interpretation:
 - Error term is *mean-independent* of treatment
- Implication:
 - Regression of Y_i on D_i identifies the causal effect

Student/Teacher Achievement Ratio Experiment

- **Context STAR experiment:**

- Large-scale randomized experiment (1985–1989)
- Cost: $\sim$ \$12 million
- $\sim$ 11,600 students (starting in kindergarten)

- **Objective:**

- Estimate the causal effect of class size on student outcomes

- **Treatments:**

- Small classes: 13–17 students
- Regular classes: 22–25 students
- Regular classes + teacher's aide (part-time or full-time)

- **Design:**

- Random assignment of students to treatments
- Schools participated if ≥ 3 classes per grade
- Followed students from kindergarten to grade 3

Conditional Independence Assumption (CIA)

- In RCTs:
 - Treatment is assigned randomly
- In observational studies:
 - Treatment is determined by choice $\Rightarrow$ potential selection bias
- **CIA (Unconfoundedness):**

$$\{Y_i(1), Y_i(0)\} \perp D_i \mid \mathbf{X}_i$$

- Conditional on pre-treatment covariates $\mathbf{X}_i$:
 - Treatment assignment is “as good as random”
- **Overlap (positivity):**
 - $0 < P(D_i = 1 \mid \mathbf{X}_i) < 1$

Conditional independence assumption (CIA)

Note that under the CIA,

$$\begin{aligned}\text{ATE} &= \mathbb{E}[Y_i(1)] - \mathbb{E}[Y_i(0)] \\ &= \mathbb{E}_{\mathbf{X}} \{ \mathbb{E}[Y_i(1) \mid \mathbf{X}_i] - \mathbb{E}[Y_i(0) \mid \mathbf{X}_i] \} \\ &= \mathbb{E}_{\mathbf{X}} \{ \mathbb{E}[Y_i(1) \mid \mathbf{X}_i, D_i = 1] - \mathbb{E}[Y_i(0) \mid \mathbf{X}_i, D_i = 0] \} \\ &= \mathbb{E}_{\mathbf{X}} \{ \mathbb{E}[Y_i \mid \mathbf{X}_i, D_i = 1] - \mathbb{E}[Y_i \mid \mathbf{X}_i, D_i = 0] \} \\ &= \mathbb{E}_{\mathbf{X}} \{ \tau(\mathbf{X}_i) \} ,\end{aligned}$$

and

$$\tau(\mathbf{X}_i) = \mathbb{E}[Y_i \mid \mathbf{X}_i, D_i = 1] - \mathbb{E}[Y_i \mid \mathbf{X}_i, D_i = 0]$$

is the *conditional average treatment effect* (CATE) at covariate value $\mathbf{X}_i$, which captures treatment effect heterogeneity across different values of $\mathbf{X}$.

Conditional independence assumption (CIA)

Under the CIA, the causal effect of D on Y can be identified by conditioning on $\mathbf{X}$, as this set satisfies the *back-door criterion*.

Back-door criterion:

- No variable in $\mathbf{X}$ is a descendant of D
- $\mathbf{X}$ blocks all back-door paths from D to Y

Back-door path:

- Any path from D to Y that starts with an arrow pointing into D

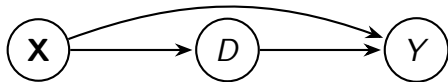


Figure 3: Directed Acyclic Graph (DAG) implied by the Conditional Independence Assumption (CIA).

Conditional Independence Assumption (CIA)

- Extend to multiple regression by including pre-treatment covariates:

$$Y_i = \beta_0 + \tau D_i + \mathbf{X}_i^\top \boldsymbol{\beta} + \mu_i$$

- In RCTs:
 - $D_i \perp \mathbf{X}_i$
 - $\Rightarrow$ identification of τ is unaffected
- Why include $\mathbf{X}_i$?
 - Explains variation in Y_i
 - $\Rightarrow$ improves precision of $\hat{\tau}$

Example: 401(k) Eligibility and Financial Assets

- **Objective:**

- Estimate the causal effect of 401(k) eligibility ($e401$)
- Outcome: net financial assets (net_tfa)

- **Identification strategy:**

- Eligibility is *conditionally exogenous*
- After controlling for job-related characteristics

- **Controls (X):**

- Age, income, family size, education
- Marital status, two-earner household
- Defined benefit pension (db), IRA participation ($pira$)
- Home ownership ($hown$)

- **Key assumption (CIA):**

$$e401 \perp net_tfa \mid \mathbf{X}$$

Bad Controls and Collider Bias

- Identification relies on conditioning on **X**
- **Warning:** more controls are not always better
 - Some variables are *bad controls* (Angrist and Pischke, 2009)
- **Collider:**
 - A variable caused by two (or more) variables
 - Example: $D \rightarrow C \leftarrow \mathbf{X}$
- **Key insight:**
 - Colliders *block* paths by default
 - Conditioning on a collider *opens* the path
- **Implication:**
 - Conditioning on C induces a spurious link:

$$D \leftrightarrow \mathbf{X} \rightarrow Y$$

- $\Rightarrow$ biased estimate of the causal effect

Collider Bias (DAG)

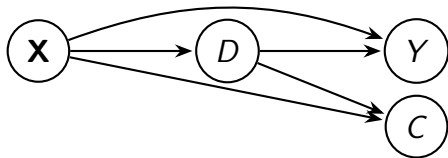


Figure 4: Directed Acyclic Graph (DAG): The additional collider *C* caused by both *D* and **X** would open a path between *D* and **X**, inducing bias.

Conditional independence assumption (CIA): Birth-weight “paradox”, collider bias

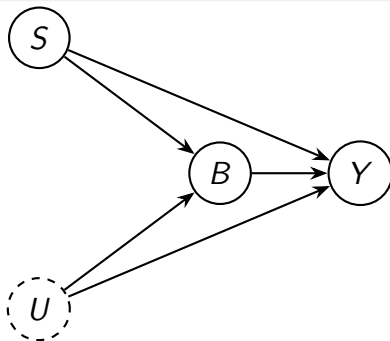


Figure 5: DAG illustrating the birth-weight paradox: S (smoking), B (birth weight), Y (infant mortality), and U (other unobserved health factors).

Instrumental variables

- *Relevance*, meaning the instruments are correlated with the treatment:

$$\mathbf{Z}_i \not\perp D_i.$$

- *Exogeneity*, which in the linear model requires:

$$\mathbb{E}[\mu_i \mid \mathbf{Z}_i] = 0,$$

and, in terms of potential outcomes, entails:

- *Exclusion restriction*: instruments affect the outcome only through the treatment:

$$Y_i(D_i = d, \mathbf{Z}_i = \mathbf{z}) = Y_i(D_i = d, \mathbf{Z}_i = \mathbf{z}') = Y_i(D_i = d), \\ d \in \{0, 1\},$$

- *Marginal exchangeability*: instruments are independent of the potential outcomes:

$$\mathbf{Z}_i \perp \{Y_i(1), Y_i(0)\}.$$

Instrumental Variables: Monotonicity

- Instrument validity conditions alone imply *partial identification* of LATE
 - See (Manski, 1989, 1990, 1995, 2003)
- To achieve **point identification**, we impose:
- **Monotonicity**
 - The instrument shifts treatment decisions in the same direction for all units

$$D_i(1) \geq D_i(0), \quad i = 1, \dots, N$$

- (No “defiers” when $Z_i \in \{0, 1\}$)

Instrumental variables

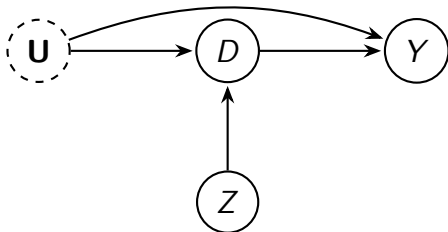


Figure 6: Directed Acyclic Graph (DAG) showing an instrumental variable Z used to identify the causal effect of D on Y when some confounders U (dashed) are unobserved or measured with error.

Instrumental variables

Table 1: Distribution of compliance types by instrument and treatment

Instrument	Treatment	
	$D_i = 0$	$D_i = 1$
$Z_i = 0$	Compliers, Never-takers	Always-takers
$Z_i = 1$	Never-takers	Compliers, Always-takers

Instrumental variables

$$\begin{aligned}\tau_{LATE} &= \mathbb{E}[Y_i(1) - Y_i(0) \mid D_i(1) = 1, D_i(0) = 0] \\ &= \frac{\mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0]}{\mathbb{E}[D_i \mid Z_i = 1] - \mathbb{E}[D_i \mid Z_i = 0]}.\end{aligned}$$

- The IV estimand is:
 - Ratio of the effect of Z on Y (reduced form)
 - To the effect of Z on D (first stage)
- Interpretation:
 - Numerator: how outcomes respond to the instrument
 - Denominator: how treatment responds to the instrument
- Key insight:
 - The instrument induces exogenous variation in D
 - $\Rightarrow$ isolates the causal effect for *compliers*

Instrumental variables: A simple setting for point and set identified LATE

Table 2: Distribution of units by type, instrument, treatment, and outcomes

Type	Z_{obs}	D_{obs}	Y_{obs}	Units
Compliers, Never-takers	0	0	3	50
Never-takers	1	0	4	20
Compliers, Always-takers	1	1	7	100
Always-takers	0	1	5	30

Example: 401(k) Participation and Financial Assets

- **Objective:**
 - Estimate the causal effect of 401(k) participation (*p401*)
 - Outcome: net financial assets (*net_tfa*)
- **Endogeneity concern:**
 - Participation in a 401(k) plan is not random. Individuals who participate may differ systematically in saving preferences
- **Instrument:** eligibility for a 401(k) plan (*e401*)
- **Controls (X):**
 - Age, income, family size, education, home ownership (*hown*), marital status, two-earner household, defined benefit pension (*db*), IRA participation (*pira*)
- **IV assumptions:**
 - *Relevance:* *e401* affects 401(k) participation
 - *Exogeneity:* *e401* affects net financial assets only through participation

Difference-in-Differences (2×2)

- Setup:
 - Two periods: $t = 1$ (pre), $t = 2$ (post)
 - Two groups:
 - Treated: $D_i = 1$ (treated between $t = 1$ and $t = 2$)
 - Control: $D_i = 0$ (never treated)
- Data:
 - Outcomes Y_{it} for $i = 1, \dots, N$
 - Balanced panel
- Potential outcomes:
 - $Y_{it}(1)$: treated in period 2
 - $Y_{it}(0)$: never treated
- **Target parameter (ATT):**

$$\tau_2 = \mathbb{E}[Y_{i2}(1) - Y_{i2}(0) \mid D_i = 1]$$

Difference-in-Differences (2×2)

- *Parallel trends:*

$$\mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid D_i = 1] = \mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid D_i = 0],$$

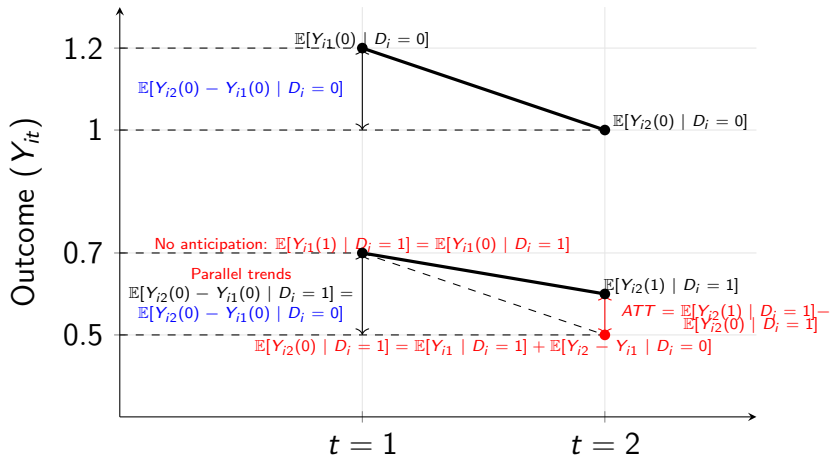
which states that, in the absence of treatment, the average change in outcomes for the treated and control groups would have evolved similarly over time.

- *No anticipation effects:*

$$\mathbb{E}[Y_{i1}(0) \mid D_i = 1] = \mathbb{E}[Y_{i1}(1) \mid D_i = 1],$$

meaning that the treatment has no causal effect prior to its implementation.

Difference-in-Differences (2×2)



Difference-in-Differences (2×2)

The ATT is the difference in outcome differences between treated and control units,

$$\tau_2 = \mathbb{E}[Y_{i2} - Y_{i1} \mid D_i = 1] - \mathbb{E}[Y_{i2} - Y_{i1} \mid D_i = 0], \quad (1)$$

Difference-in-Differences (2×2)

En regression forms:

$$Y_{it} = \alpha_i + \phi_t + \tau_2 D_i + \mu_{it},$$

which implies

$$Y_{i2} - Y_{i1} = (\phi_2 - \phi_1) + \tau_2 D_i + (\mu_{i2} - \mu_{i1}).$$

The two-way fixed effects (TWFE) model

$$Y_{it} = \alpha + \alpha_i + \phi_t + \tau_2 [D_i \cdot \mathbb{1}(t = 2)] + \epsilon_{it},$$

where $\mathbb{1}(t = 2)$ is an indicator for the post-treatment period (Roth et al., 2023).

Difference-in-Differences (2×2)

- Extension: condition on covariates $\mathbf{X}$
 - Compare units that are similar in observed characteristics
 - Identification assumptions hold *conditional on* $\mathbf{X}$
- Estimation:
 - Linear regression (parametric)
- Alternatives (more flexible):
 - *Outcome regression* (OR)
 - *Inverse probability weighting* (IPW)
 - *Doubly robust* (OR + IPW)
- Key advantage:
 - Consistent under conditional parallel trends
 - No need to assume a linear CEF

Difference-in-Differences (2×2): Assumptions

- **Functional form matters:**
 - Parallel trends may not be preserved under transformations of Y
- **Limitations:**
 - Parallel trends and no anticipation are *not testable*
 - They involve counterfactual outcomes we do not observe
- **What can we do?**
 - Examine pre-treatment trends
 - Assess plausibility of the identifying assumptions

Staggered DiD (Multiple Periods)

- Many applications go beyond the 2×2 design
- Key idea:
 - Staggered DiD is an aggregation of multiple 2×2 comparisons
 - Compare treated units with *not-yet-treated* units (Baker et al., 2025)
- **Setup:**
 - Periods: $t = 1, \dots, T$
 - Treatment is *absorbing*: once treated, always treated

$$G_i = \min\{t : D_{it} = 1\}$$

- G_i : first period in which unit i is treated

Staggered DiD (Multiple Periods)

- **Potential outcomes:**

- $Y_{it}(g)$: outcome for a unit first treated in period g
- $Y_{it}(\infty)$: outcome for a never-treated unit

$$ATT(g, t) = \mathbb{E}[Y_{it}(g) - Y_{it}(\infty) \mid G_i = g]$$

- $ATT(g, t)$: average treatment effect for units first treated in period g , evaluated at time t
- Each cohort g has its own treatment effects over time

Staggered DiD: Identifying Assumptions

- 1. Staggered parallel trends (not-yet-treated):

$$\mathbb{E}[Y_{it}(\infty) - Y_{it-1}(\infty) \mid G_i = g] = \mathbb{E}[Y_{it}(\infty) - Y_{it-1}(\infty) \mid G_i = g'],$$

- Interpretation:
 - In the absence of treatment, groups would evolve in parallel
 - Comparison group: *not-yet-treated* units
- Alternative comparison groups:
 - Never-treated*: $g' = \infty$
 - All groups*: parallel trends across all g, g'

Staggered DiD: Additional Assumptions

- **2. No anticipation:**

$$\mathbb{E}[Y_{it}(g)] = \mathbb{E}[Y_{it}(\infty)], \quad \forall t < g$$

- Units do not change behavior before treatment

- **Trade-offs across assumptions:**

- *All groups*: stronger assumption, more precision
- *Never-treated*: weaker assumption, less precision
- *Not-yet-treated*: intermediate case

- **Implication:**

- Long-run effects: require parallel trends after treatment
- Short-run effects: require parallel trends in the relevant period

Staggered DiD: Identification and Dynamics

- Under the identifying assumptions, $ATT(g, t)$ can be written as:

$$ATT(g, t) = \mathbb{E}[Y_{it} - Y_{i,g-1} \mid G_i = g] - \mathbb{E}[Y_{it} - Y_{i,g-1} \mid G_i = g'],$$

- Interpretation:
 - Change in outcomes for cohort g
 - Relative to a comparison group (not-yet-treated or never-treated)
- Key features:
 - Cohorts indexed by first treatment period g
 - Effects evaluated at time t
 - $\Rightarrow$ multiple cohort-time effects $ATT(g, t)$
- **Dynamics:**
 - Use *event-study* plots
 - Track effects before and after treatment

Staggered DiD: Placebo Tests

- Advantage of multiple pre-treatment periods:

- Enables *placebo (falsification) tests*

- Under **no anticipation**:

- Treatment has no effect before it occurs

$$\mathbb{E}[Y_{it} - Y_{ib} \mid G_i = g] = \mathbb{E}[Y_{it} - Y_{ib} \mid G_i = g'], \quad b < t < \min\{g, g'\}$$

- **Test:**

- Pre-treatment differences should be statistically zero

- **Interpretation:**

- Supports (but does not prove) parallel trends

- **Caution:**

- Parallel trends is fundamentally *untestable*
 - Requires sensitivity analysis

Staggered DiD with Covariates

- Assume parallel trends holds *conditional on* $\mathbf{X}_i$
- **1. Conditional parallel trends (not-yet-treated):**

$$\mathbb{E}[Y_{it}(\infty) - Y_{i,t-1}(\infty) \mid G_i = g, \mathbf{X}_i] = \mathbb{E}[Y_{it}(\infty) - Y_{i,t-1}(\infty) \mid G_i = g', \mathbf{X}_i], \\ t \geq g, g' > t$$

- **2. Overlap (positivity):**

$$P(G_i = g \mid \mathbf{X}_i) < 1 - \epsilon, \quad \epsilon > 0, \quad P(G_i = g) > 0$$

- Interpretation:
 - Comparable units exist across cohorts given $\mathbf{X}_i$

Staggered DiD with Covariates

- Under these assumptions (and no anticipation):

$$ATT(g, t) = \mathbb{E}[Y_{it} - Y_{i,g-1} \mid G_i = g] \\ - \mathbb{E}_{\mathbf{X}}\{\mathbb{E}[Y_{it} - Y_{i,g-1} \mid \mathbf{X}_i, G_i = g'] \mid G_i = g\}$$

- **Estimation:**

- Outcome regression (OR)
- Inverse probability weighting (IPW)
- Doubly robust methods

- **Caution:**

- TWFE does *not* generally identify $ATT(g, t)$
- Can produce non-convex weights (even negative)

Application: Minimum Wage and Teen Employment (Callaway and Sant'Anna, 2021)

- **Context:**

- Period: 2001–2007 (federal minimum wage fixed at \$5.15)
- Outcome: county-level teen employment

- **Treatment definition:**

- States that increased minimum wage $\Rightarrow$ treated
- Groups defined by first year of increase (staggered adoption)
- States with no increase $\Rightarrow$ control group

- **Why this design?**

- More variation than local case studies
- Clean identification from state-level policy changes

Application: Data and Empirical Features

- **Data:**

- Teen employment: Quarterly Workforce Indicators (QWI)
- County characteristics:
 - Population, race, education
 - Income and poverty measures

- **Sample:**

- 29 states after exclusions

- **Empirical advantages:**

- Enables staggered DiD estimation
- Allows testing of pre-treatment trends
- Supports event-study analysis

Regression Discontinuity Design (RD)

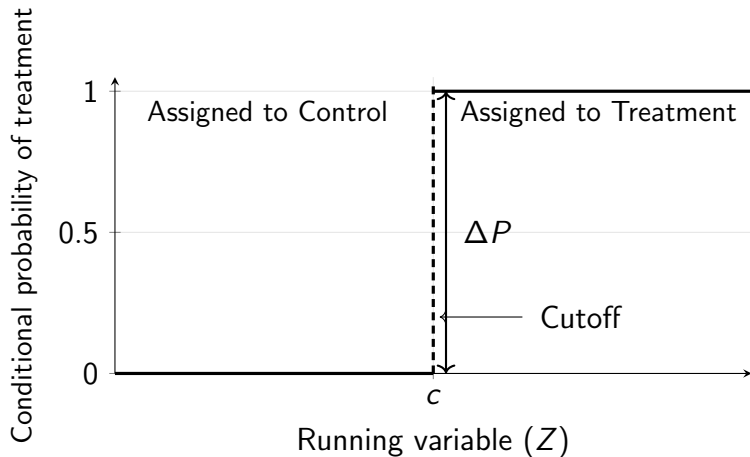
- **Key ingredient:** running variable Z_i
 - Also called forcing, index, or score variable
 - Determines treatment assignment relative to cutoff c
- **Identification idea:**
 - Potential outcomes vary *smoothly* in Z_i
 - Any discontinuity at c is attributed to treatment
- **Treatment jump at the cutoff:**

$$\begin{aligned}\Delta &= \lim_{z \downarrow c} \Pr(D_i = 1 \mid Z_i = z) - \lim_{z \uparrow c} \Pr(D_i = 1 \mid Z_i = z) \\ &= \begin{cases} 1, & \text{sharp RD,} \\ \in (0, 1), & \text{fuzzy RD} \end{cases}\end{aligned}$$

- Right and left limits: approach c from above and below

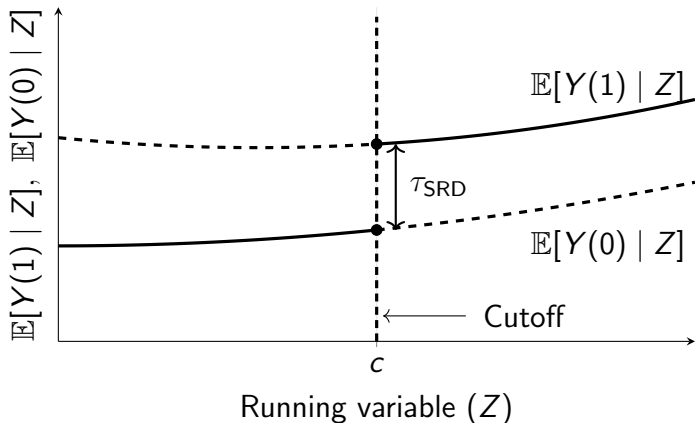
Regression discontinuity design (RD): Sharp

$$D_i = \mathbb{1}\{Z_i \geq c\},$$



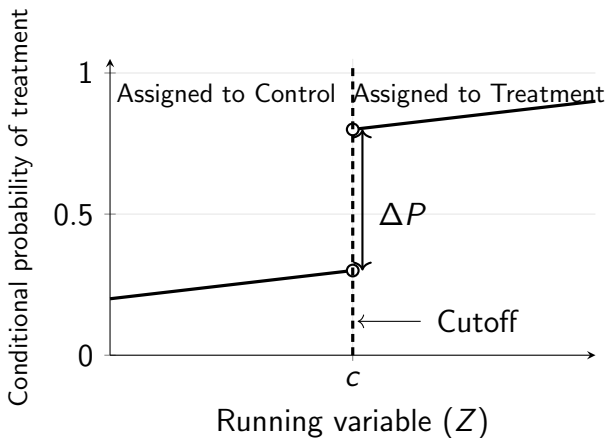
Regression discontinuity design (RD): Sharp

$$\tau_{\text{SRD}} = \lim_{z \downarrow c} \mathbb{E}[Y_i | Z_i = z] - \lim_{z \uparrow c} \mathbb{E}[Y_i | Z_i = z],$$



Regression discontinuity design (RD): Fuzzy

$$0 < \lim_{z \downarrow c} \Pr(D_i = 1 \mid Z_i = z) - \lim_{z \uparrow c} \Pr(D_i = 1 \mid Z_i = z) < 1.$$



Regression discontinuity design (RD): Fuzzy

The identification assumptions are:

1. *Continuity of the potential outcome regression functions*, particularly at c .

2. *A nontrivial first stage*,

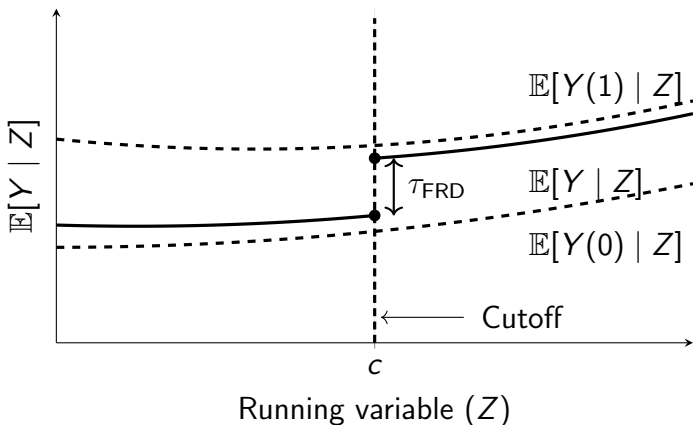
$$\lim_{z \downarrow c} \mathbb{E}[D_i \mid Z_i = z] - \lim_{z \uparrow c} \mathbb{E}[D_i \mid Z_i = z] \neq 0,$$

and

3. *Monotonicity*: $D_i(z)$ is non-increasing in z at $z = c$.

Regression discontinuity design (RD): Fuzzy

$$\tau_{\text{FRD}} = \frac{\lim_{z \downarrow c} \mathbb{E}[Y_i | Z_i = z] - \lim_{z \uparrow c} \mathbb{E}[Y_i | Z_i = z]}{\lim_{z \downarrow c} \mathbb{E}[D_i | Z_i = z] - \lim_{z \uparrow c} \mathbb{E}[D_i | Z_i = z]}.$$



Application: RDD and Close Elections

- **Context:** (Meyersson, 2014)
 - Study the effect of Islamic mayors in Turkey
 - Outcome: women's educational attainment
- **Running variable:**
 - Vote share of Islamic party candidates
 - Cutoff: winning vs losing the election
- **Treatment:**
 - $D_i = 1$: Islamic mayor elected
 - $D_i = 0$: secular mayor elected
- **Key idea (RDD):**
 - Close elections $\Rightarrow$ quasi-random assignment
 - Compare municipalities just above and below the cutoff

Application: Why RDD?

- **Challenge:**
 - Municipalities differ systematically
 - Example: religious conservatism
- **Problem with naive comparison:**
 - Islamic vs secular areas are not comparable
 - Differences may affect women's outcomes
- **RDD solution:**
 - Focus on close elections (around cutoff)
 - Units near the cutoff are similar
 - $\Rightarrow$ isolates causal effect of Islamic mayor

Application: Outcome and Interpretation

- **Outcome:**

- Women's educational attainment
- Measured as:
 - % of women aged 15–20 (in 2000) who completed high school

- **Key feature:**

- Focus on close elections around the cutoff
- $\Rightarrow$ quasi-random assignment of Islamic vs secular mayors

- **Interpretation:**

- Identifies the causal effect of Islamic rule
- For municipalities near the electoral threshold

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